Knee Surg Sports Traumatol Arthrosc (2013) 21:576–583
DOI 10.1007/s00167-012-1947-0

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Long-term follow-up of patellar tendon grafts or hamstring tendon grafts in endoscopic ACL reconstructions

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Received: 5 October 2011 / Accepted: 27 February 2012 / Published online: 10 March 2012

Abstract
Purpose Several studies compare the short- and long-term results of anterior cruciate ligament (ACL) reconstruction using bone-patellar tendon-bone (BPTB) graft or double-looped semitendinosus and gracilis (DLSG) graft. However, no studies evaluate the long-term results of BPTB grafts fixed with metal interference screws and DLSG grafts fixed with the Bone Mulch Screw and the Washer Loc. This prospective randomized multicentre study has the null hypothesis that there is no difference in long-term outcome between the two procedures.
Methods A total of 114 patients with a symptomatic ACL rupture were randomized to reconstruction with either a BPTB graft ($N = 58$) or a DLSG graft ($N = 56$). Follow-up was conducted after one, two and seven years. At the seven-year follow-up, 102 of the 114 patients (89%) were available for evaluation; however, 16 of these by telephone-interview only.
Results Ten patients in the BPTB group and 19 patients in the DLSG group underwent additional knee surgery ($P = 0.048$), two and three, respectively, of these were ACL revisions (n.s.). The total flexion work was lower in the DLSG group ($P = 0.001$). The mean peak flexion torque and extension work, however, showed no difference between the groups. No significant differences were found between the groups regarding the Tegner activity score, the Lysholm functional score, the Knee injury and osteoarthritis outcome score (KOOS), subjective knee function, anterior knee pain or mobility. There was no significant difference in laxity between the groups on the Lachman test or the KT-1,000 maximum manual force test.
Conclusions Both grafts and fixation methods resulted in satisfactory subjective outcome and objective stability. Both these methods can therefore be considered as suitable alternatives for ACL reconstructions.
Level of evidence II.

Keywords Anterior cruciate ligament (ACL) reconstruction · Patellar tendon · Hamstring tendon · Autograft · Prospective · Randomized · Long-term

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Introduction

Patients with a rupture of the anterior cruciate ligament (ACL) may report pain, difficulty with athletic performance and/or giving-way symptoms in daily activities [3, 10, 27]. ACL rupture may lead to knee laxity, resulting in functional instability and increased risk of meniscal injury [6, 20] and degenerative joint disease [12, 35]. The number of ACL reconstructions was found to be between 30 and 40 per 100,000 inhabitants in the Scandinavian countries [14, 23]. The corresponding incidence increased to 85 per 100,000 for the 16–39 years age group in Norway [14]. This is not surprising as sport activities are popular at this age. Women are reported to have a five times higher risk of having ACL rupture compared to men [26]. The bone-patellar tendon-bone (BPTB) graft and the double-looped semitendinosus and gracilis (DLSG) graft are two of the most common grafts used for ACL surgery. Kartus et al. [19] state in their review study that 40–60% of patients report various donor-site problems after harvesting of BPTB grafts. Such problems include anterior knee pain, decreased sensitivity of the knee and kneeling pain. With the use of DLSG grafts, less donor-site complications have been reported [24, 28], however, prominence or pain from the hardware [13], femoral tunnel widening [2, 11], increased knee laxity [11, 13] and weakness of the hamstring muscles postoperatively [4, 24] have been described. Satisfactory stability of the knees has been reported with both procedures [2, 8, 21, 22, 24, 32].

The aim of this study was to compare the seven-year follow-up results after ACL reconstructions with either BPTB grafts fixed with metallic interference screws or DLSG grafts fixed with Bone Mulch Screw on the femur and WasherLoc on the tibia. The null hypothesis of this prospective randomized multicentre study was that there were no differences in the seven-year clinical results between the two groups. To our knowledge, there are no previous studies comparing these fixation methods.

Materials and methods

One-hundred-and-fourteen patients, 72 men and 42 women, with a median age of 27 years (range, 18–49) with a symptomatic tear of the ACL were randomized to endoscopic reconstruction with either a BPTB graft ($N = 58$) or a DLSG graft ($N = 56$) (Drogset et al. [7] presented a total number of 115 patients included when presenting the first follow-up results. When reviewing the material, we were not able to find inclusion-papers for more than 114 patients; hence, this is the number of patients considered in this study. This fact also results in a small age difference between the two studies). The surgery was carried out at four different hospitals between 2001 and 2004. The study was approved by The Regional Ethics Committee. All patients received written and oral information about the project and signed a written consent form. Patients between 18 and 50 years with an isolated symptomatic ACL rupture and two plane radiographs of the knee were included. The exclusion criteria were major additional injury of the knee, the need for major meniscal repairs, cartilage injuries (Outerbridge grade 3–4), malalignment with more than 5° varus or valgus or ACL injuries in the opposite knee [7]. The patients were randomized preoperatively with the aid of sealed envelopes, to one of the two surgical procedures. At the seven-year follow-up, patients completed questionnaires with an evaluation of subjective knee function, the Lysholm functional score [36], Tegner activity score [36] and Knee injury and osteoarthritis outcome score (KOOS) [30]. Additional information concerning subsequent injuries or reoperations of the knee was also noted. Two independent observers performed full clinical knee-examinations including range of motion, swelling, the Lachman test and the KT-1000 maximum manual force test. Isokinetic testing of muscle strength was carried out with a Biodex 6000 dynamometer® (Biodex Medical Systems Inc., Shirley, NY, USA) by experienced physiotherapists.

Surgical technique and rehabilitation

The procedures were carried out under spinal or general anaesthesia with the use of a tourniquet. If necessary, treatment of the menisci and/or a notchplasty was performed. The BPTB grafts were fixed with metallic interference screws (Linvatec, Largo, Florida, USA), while the Bone Mulch Screw® and the WasherLoc® (Biomet, Inc. Warsaw, Indiana, USA) were used for fixation of the DLSG grafts. Postoperatively the patients in both groups followed the same intensive rehabilitation programme including immediate knee movement exercises, full passive extension several times a day and early weight-bearing. No brace was used. Sport activities were gradually allowed. The operative techniques for both grafts and the rehabilitation programme were described in detail in a previous study [7].

Statistical analysis

Statistical methods used have previously been reported in detail [7]. Statistical analysis was conducted with the SPSS software package, version 16.0. The t test for equality of means was used for comparison of differences between groups and the chi-square test was used for categorical variables. Differences were considered statistically significant when $P < 0.05$.

Results

Seventy-seven of 114 patients (68%, 41 in the BPTB group and 36 in the DLSG group) were available for clinical examination at the median seven-year follow-up (range, 63–94 months). In addition, 16 patients (seven in the BPTB group and nine in the DLSG group) agreed to a telephone-interview including questions regarding subjective knee function, concomitant surgery, Tegner activity score and Lysholm functional score. Five patients (two in the BPTB group and three in the DLSG group) were excluded from further examination and analysis because of ACL re-rupture, and four patients (one in the BPTB group and three in the DLSG group) were excluded because of ACL injury in the opposite knee. Twelve patients could not be traced (Fig. 1). During the follow-up period, 10 patients (20%) in the BPTB group and 19 patients (37%) in the DLSG group underwent subsequent knee surgery (P = 0.048) (Table 1). Nine patients in the DLSG group and six in the BPTB group underwent meniscus surgery (n.s.).

Muscle strength was measured in 33 patients in the DLSG group and 37 patients in the BPTB group (Fig. 2). After seven years, the total flexion work values in the injured knee, compared to the uninjured knee were increased by 2.4 N/m (SD ± 64, range −154 to 134) in the BPTB group and reduced by 61 N/m (SD ± 81, range −206 to 119) in the DLSG group ($P = 0.001$). This gave a 12% reduction in total hamstring work in the DLSG group. The mean peak flexion torque, compared to the uninjured knee, was reduced by 4.5 N/m (SD ± 28, range −160 to 21) in the BPTB group and 8.0 N/m (SD ± 15, range −42 to 29) in the DLSG group (n.s.). The extension work measurements showed no differences between the groups. An extension deficit of five degrees or more was observed in three patients (8%) in the DLSG group and two patients (5%) in the BPTB group (n.s.). A flexion deficit of five degrees or more was observed in six patients (17%) in the DLSG group and four patients (10%) in the BPTB group (n.s.). One patient (3%) in the DLSG group and four (10%) in the BPTB group had pain on palpating the lower pole of the patella at the seven-year follow-up (n.s.).

Twelve patients (33%) in the DLSG group and 15 patients (37%) in the BTPB group had a Lachman test grade one (n.s.). One patient in each group had a Lachman test grade two. The mean side-to-side difference on the KT-1000 maximum manual force test was 1.4 mm (SD ± 1.4, range −2 to 5) in the DLSG group; a side-to-side difference of more than 3 mm was observed in three of the patients. In the BPTB group, the mean side-to-side difference was also 1.4 mm (SD ± 1.8, range −5 to 4); two of the patients had greater than 3 mm side-to-side difference. There was no significant difference between the groups regarding the Lachman test and the KT-1000 maximum manual force test. Thirty-two patients (71%) in the DLSG group and 35 patients (73%) in the BPTB group classified their subjective knee function as excellent or

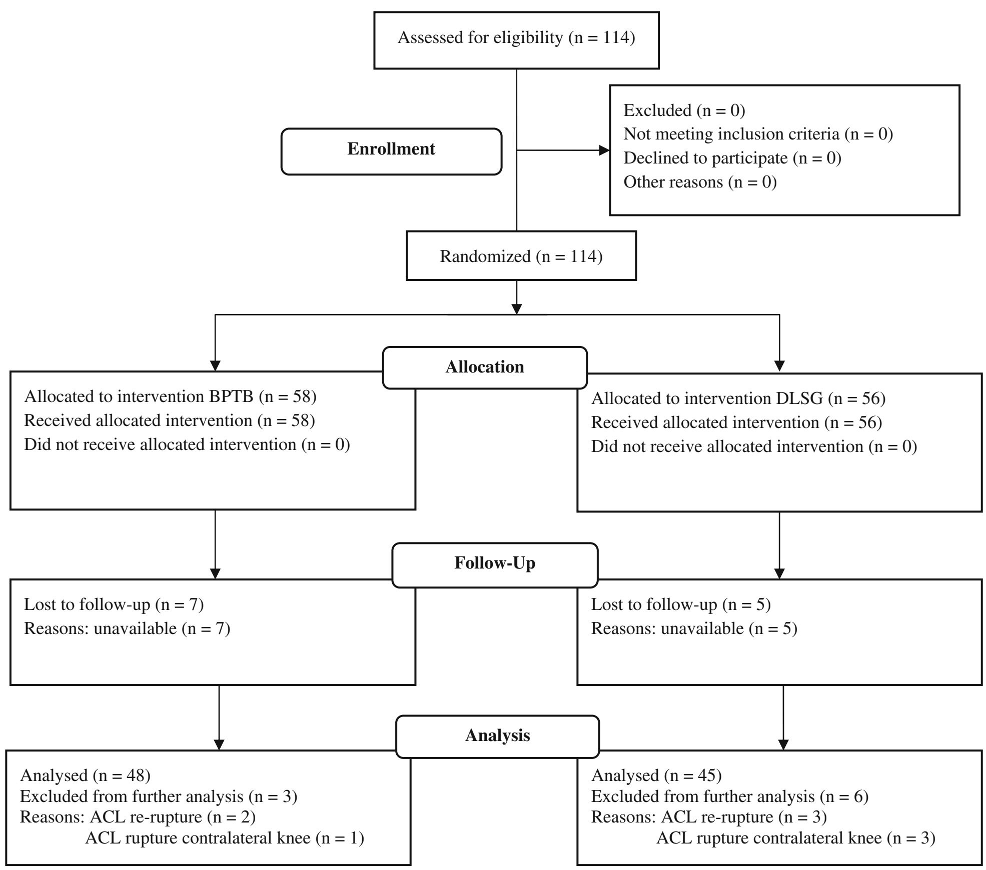

Fig. 1 Patient flowchart

Table 1 Comparison of subsequent knee surgery at the seven-year follow-up

	DLSG	BPTB
Med. meniscus resections	6	4
Lat. meniscus resections	3	1
Med. meniscus repair	0	1
Lat. meniscus repair	0	0
ACL revisions	3	2
Notchplasty	0	1
Debridement	3	1
Synovectomy	1	0
Cartilage surgery	1	0
Other surgery	2	0

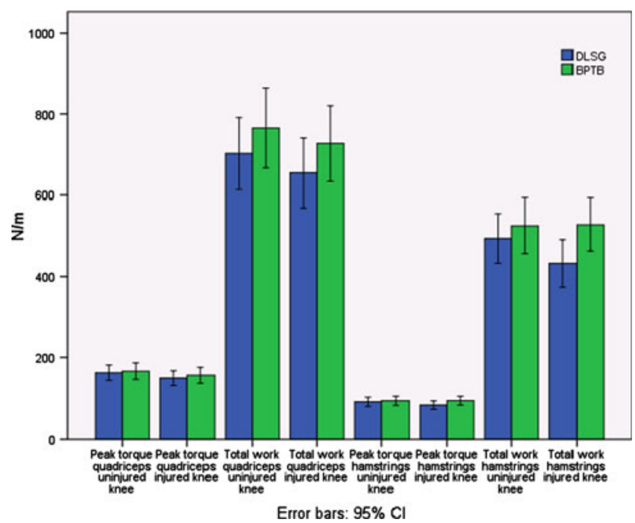

Fig. 2 The strength measurements in the two groups at the seven-year follow-up

good (Fig. 3). There were no statistically significant differences between the groups on either subjective knee function, Tegner score (Fig. 4), Lysholm score (Fig. 5) or KOOS (Fig. 6).

Discussion

The main findings of this seven-year follow-up were that both grafts show satisfactory results, and that differences in symptoms and findings present at the two-year follow-up [7] become less noticeable with time. Autogenous patellar tendon grafts and hamstring tendon autografts are the two most common grafts used for ACL reconstructions. There have been studies comparing the clinical outcome of these two grafts [9, 11, 21, 22, 34], but to our knowledge, there is no other prospective randomized study comparing long-term results for BPTB graft and DLSG graft with the

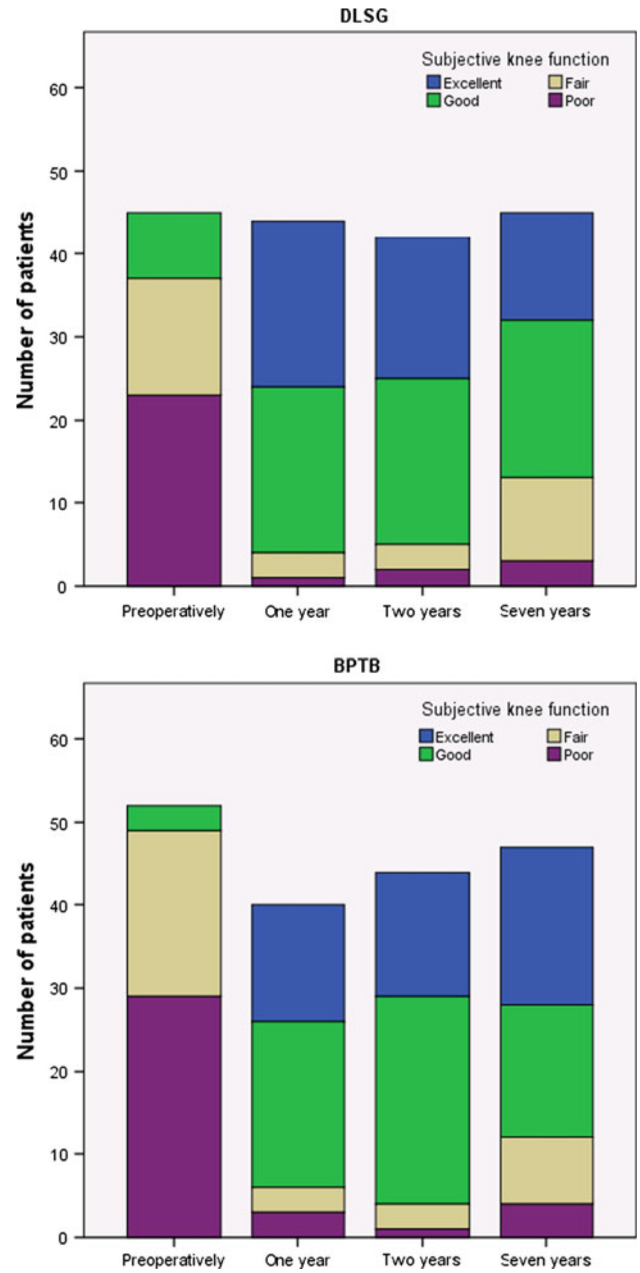

Fig. 3 Subjective knee function in the two groups preoperatively and at one-, two- and seven-year follow-up

fixation methods described above. The design of this study is one of its strengths.

Freedman et al. [13] concluded that ACL reconstruction using BPTB graft had a lower rate of graft failures, improved static stability and increased patient satisfaction compared with the DLSG graft. Yunes et al. meta-analysis [40] also found better stability and a higher postoperative activity level in the BPTB group compared to the patients in the DLSG group. Other sources [5, 11, 25] also report more stable knees after BPTB reconstructions. Wilson et al. [38] in a biomechanical analysis compared load to

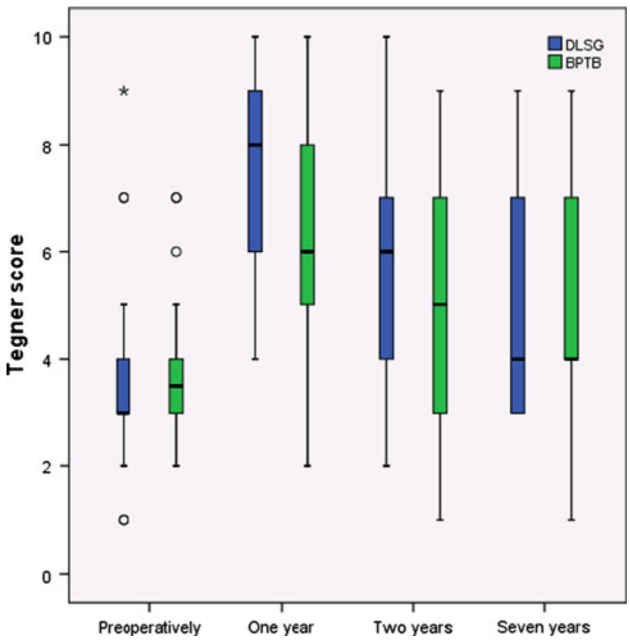

Fig. 4 Median Tegner score in the two groups preoperatively and at one-, two- and seven-year follow-up

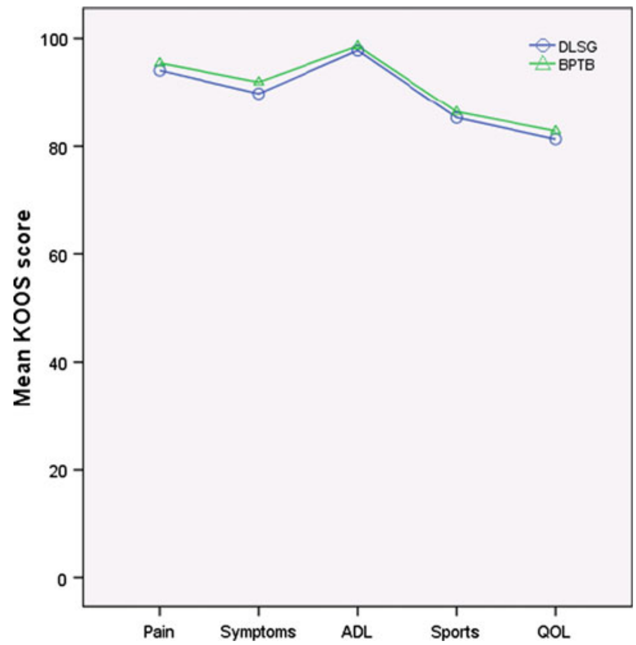

Fig. 6 Mean KOOS score in the two groups at the seven-year follow-up

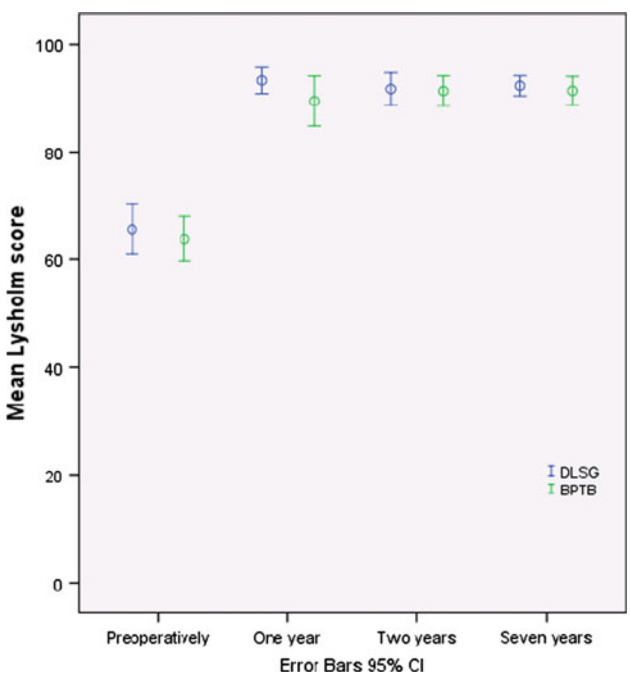

Fig. 5 Mean Lysholm score in the two groups preoperatively and at one-, two- and seven-year follow-up

failure and graft stiffness of matched BPTB grafts with DLSG graft. They found that the DLSG grafts were significantly stronger than the BPTB grafts; the stiffness values, however, were similar [38]. In the present study, we did not find any significant difference in re-rupture rate between the groups; two patients in the BPTB group (4%) and three in the DLSG group (6%). At a five-year follow-up, Sajovic et al. [32] found an ACL revision rate of 8% in the BPTB group and 7% in the DLSG group (n.s.). Pinczewski et al. [28] reported respectively 8% and 13% revisions (n.s.) 10 years after the surgery. During the follow-up period in the present study, there was a significantly higher frequency of patients having subsequent knee surgery in the DLSG group (37%) compared to the BPTB group (20%) ($P = 0.048$). In Freedman et al. meta-analysis [13], like in the present study, there was no significant difference between the two groups regarding the prevalence of subsequent meniscal surgery. We did not find any significant difference between the groups regarding contralateral ACL rupture; one rupture in the BPTB group (2%) and three in the DLSG group (6%). In contrast, Pinczewski et al. [28] reported 20 contralateral ruptures (22%) in the BPTB group and nine (10%) in the DLSG group ($P = 0.02$) after 10 years.

In our study, muscle strength in the injured knee compared to the uninjured knee was statistically significantly lower concerning total flexion work in the DLSG group ($P = 0.001$), similar to the one- and two-year follow-up [7]. Some studies show similar results [4, 24, 25], while others find no significant differences [2, 17, 18]. The reduced hamstring strength found among those with DLSG grafts may have affected the function of the knee and made them more vulnerable to new traumas. This could be one explanation as to why a larger need of subsequent knee surgery in the DLSG group compared to the BPTB group was found. The peak flexion torque was significantly lower in the DLSG group after one year; however, no difference was found after two [7] and seven years. Aune et al. [4] suggested that hamstring weakness might be due to inadequate rehabilitation in the DLSG group, and that this group should focus specifically on strengthening the hamstring muscles, also after returning to sports. A possible need for slower and more specific hamstring rehabilitation was also reported in two studies by Heijne and Werner [15, 16], and this is currently under consideration at our department. In the present study, there was no difference between the groups in extension work at any time. Based on clinical experience among the senior authors, bracing was not found beneficial and was therefore not used postoperatively. A recent review [37] concluded that postoperative mobilization without bracing is preferable.

Several studies have shown that the use of BPTB grafts result in a higher rate of donor-site morbidity [2, 8, 25, 28, 39], and increased risk of anterior knee pain was one of the conclusions in Freedman et al. meta-analysis [13]. Pinczewski et al. [28] compared BPTB graft and DLSG graft at 10-year follow-up and concluded that harvest-site symptoms and kneeling pain were significantly more common in the BPTB group. In contrast, Lidèn et al. [22] and Sajovic et al. [31, 32], with respectively seven-year follow-up and five- and 11-year follow-up, did not find significant differences between patellar tendon and hamstring tendon grafts. Samuelsson et al. [33] also concluded in their review that differences in donor-site morbidity disappear with time. At two years, Drogset et al. [7] found that 12 patients in the BPTB group and four in the DLSG group had pain on palpation of the lower pole of the patella ($P = 0.04$). At the seven-year follow-up, only four patients (10%) in the BPTB group and one in the DLSG group (3%) described pain on this examination (n.s.). Aglietti et al. [1] found that an extension loss of $\leq 3°$ was noted in 47% in the BPTB group compared to 3% in their DLSG group. Other sources [11, 13] have also reported more frequent loss of extension for BPTB grafts, and one investigator has suggested that motion problems may be a consequence of increased stability [13]. There was no difference between the two groups in range of motion or anterior–posterior laxity measured with both the KT-1000 maximum manual force test and the Lachman test in the present study.

In the present long-term follow-up study, there was a trend towards less satisfactory subjective knee function with time in both groups (Fig. 3). Roe et al. [29] stated that there was no significant difference in subjective knee function between the BPTB group and DLSG group at any time, nor was there significant change over time. Heijne and Werner [16] suggested that athletes might be able to return to sports earlier and at a higher level if BTPB grafts were chosen above hamstring grafts. In contrast, Zaffagnini et al. [41] concluded that the use of double-bundle hamstring grafts gave better functional results and a faster return to sport activities. The present study did not show any difference between the groups concerning the Tegner activity score and the Lysholm functional score. Several studies report similar findings [8, 9, 18, 21, 22]. Aglietti et al. [2], as in the present study, did not find any significant difference between the two groups regarding KOOS. The use of hamstring grafts has become common for ACL reconstructions, but the data presented in the preceding sections give reason to recommend that one continues with BPTB grafts in certain cases and that one individualizes the choice of graft for each patient.

Three studies [28, 29, 32] with at least five-year follow-up report a higher rate of radiographic osteoarthritis among knees with BPTB grafts compared to DLSG grafts. The present study lacks radiographic assessments, and this represents one of the three main limitations of the study. The other two limitations are the relatively low number of patients available for clinical examination and muscle strength testing at the follow-up, and the relatively high number of surgeons participating.

Conclusion

The majority of both BPTB grafts and DLSG grafts produce excellent or good clinical outcome for ACL reconstructions at seven-year follow-up. In the DLSG group, the total flexion work was statistically significantly lower in the injured knee compared to the uninjured knee. There were also a significantly higher number of patients who underwent additional knee surgery in the DLSG group compared to the BPTB group. Anterior knee pain present in the BPTB group at two-year follow-up became less noticeable with time.

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